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3) Laburtu:

a) $x^2-3x-5=2x+9$

$x^2-3x-5-2x-9=0$

$x^2-5x-14=0 \quad \begin{cases} a=1 \\ b=-5 \\ c=-14 \end{cases}$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{-(-5)\pm\sqrt{(-5)^2-4.1.(-14)}}{2.1}=\frac{5\pm\sqrt{25+56}}{2}=$$

$$=\frac{5\pm\sqrt{81}}{2}=\frac{5\pm 9}{2}$$

$x_1=\frac{5+9}{2}=\frac{14}{2}=7 \qquad \boxed{x_1=7}$

$x_2=\frac{5-9}{2}=\frac{-4}{2}=-2 \qquad \boxed{x_2=-2}$

b) $6x^2-5(x-1)=x(x+1)+4$

$6x^2-5x+5=x^2+x+4$

$6x^2-5x+5-x^2-x-4=0$

$5x^2-6x+1=0 \quad \begin{cases} a=5 \\ b=-6 \\ c=1 \end{cases}$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{-(-6)\pm\sqrt{(-6)^2-4.5.1}}{2.5}=\frac{6\pm\sqrt{36-20}}{10}=$$

$$=\frac{6\pm\sqrt{16}}{10}=\frac{6\pm 4}{10}$$

$x_1=\frac{6+4}{10}=\frac{10}{10}=1 \qquad \boxed{x_1=1}$

$x_2=\frac{6-4}{10}=\frac{2}{10}=\frac{1}{5} \qquad \boxed{x_2=\frac{1}{5}}$

③ c) $\frac{2x^2}{1} + \frac{x}{4} = \frac{x^2}{1} + \frac{4x}{5} + \frac{1}{5}$

$$\frac{40x^2}{20} + \frac{5x}{20} = \frac{20x^2}{20} + \frac{16x}{20} + \frac{4}{20}$$

$$40x^2 + 5x - 20x^2 - 16x - 4 = 0$$

$$20x^2 - 11x - 4 = 0 \quad \begin{cases} a = 20 \\ b = -11 \\ c = -4 \end{cases}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-11) \pm \sqrt{(-11)^2 - 4.20.(-4)}}{2.20} =$$

$$= \frac{11 \pm \sqrt{121 + 320}}{40} = \frac{11 \pm \sqrt{441}}{40} = \frac{11 \pm 21}{40}$$

$$x_1 = \frac{11+21}{40} = \frac{32}{40} = \frac{4}{5}$$

$$x_1 = \frac{11-21}{40} = \frac{-10}{40} = -\frac{1}{4}$$

$$\boxed{x_1 = \frac{4}{5}} \quad \boxed{x_2 = -\frac{1}{4}}$$

d) $x(x+1) - \frac{1}{2} = \frac{x-4}{6}$

$$\frac{x^2}{1} + \frac{x}{1} - \frac{1}{2} = \frac{x-4}{6}$$

$$\frac{6x^2}{6} + \frac{6x}{6} - \frac{3}{6} = \frac{x-4}{6}$$

$$6x^2 + 6x - 3 = x - 4$$

$$6x^2 + 6x - 3 - x + 4 = 0$$

$$6x^2 + 5x + 1 = 0 \quad \begin{cases} a = 6 \\ b = 5 \\ c = 1 \end{cases}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-5 \pm \sqrt{5^2 - 4.6.1}}{2.6} = \frac{-5 \pm \sqrt{25-24}}{12} = \frac{-5 \pm \sqrt{1}}{12} =$$

$$= \frac{-5 \pm 1}{12}$$

$$x_1 = \frac{-5+1}{12} = \frac{-4}{12} = -\frac{1}{3}$$

$$x_2 = \frac{-5-1}{12} = \frac{-6}{12} = -\frac{1}{2}$$

$$\boxed{x_1 = -\frac{1}{3}} \quad \boxed{x_2 = -\frac{1}{2}}$$

③ e) $\frac{2x+2}{3} + \frac{x^2-x}{5} = \frac{3x+7}{10}$

$$\frac{10(2x+2)}{30} + \frac{6(x^2-x)}{30} = \frac{3(3x+7)}{30}$$

$$20x+20+6x^2-6x = 9x+21$$

$$6x^2+14x+20-9x-21=0$$

$$6x^2+5x-1=0 \quad \begin{cases} a=6 \\ b=5 \\ c=-1 \end{cases}$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-5 \pm \sqrt{5^2-4.6.(-1)}}{2.6} = \frac{-5 \pm \sqrt{25+24}}{12} =$$

$$= \frac{-5 \pm \sqrt{49}}{12} = \frac{-5 \pm 7}{12} \quad \begin{cases} x_1 = \frac{-5+7}{12} = \frac{2}{12} = \frac{1}{6} \\ x_2 = \frac{-5-7}{12} = -\frac{12}{12} = -1 \end{cases}$$

$$\boxed{x_1 = \frac{1}{6}} \quad \boxed{x_2 = -1}$$

f) $x^2 - \frac{1}{4} = \frac{1}{5}\left(\frac{x}{4} - 1\right)$

$$\frac{x^2}{1} - \frac{1}{4} = \frac{x}{20} - \frac{1}{5}$$

$$\frac{20x^2}{20} - \frac{5}{20} = \frac{x}{20} - \frac{4}{20}$$

$$20x^2-5-x+4=0$$

$$20x^2-x-1=0 \quad \begin{cases} a=20 \\ b=-1 \\ c=-1 \end{cases}$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-(-1) \pm \sqrt{(-1)^2-4.20.(-1)}}{2.20} = \frac{1 \pm \sqrt{1+80}}{40} =$$

$$= \frac{1 \pm \sqrt{81}}{40} = \frac{1 \pm 9}{40} \quad \begin{cases} x_1 = \frac{1+9}{40} = \frac{10}{40} = \frac{1}{4} \\ x_2 = \frac{1-9}{40} = \frac{-8}{40} = \frac{-1}{5} \end{cases}$$

$$\boxed{x_1 = 1/4} \quad \boxed{x_2 = -1/5}$$

③ g) $\frac{x}{2}\left(\frac{x}{1}+\frac{1}{30}\right)=\frac{x}{3}\left(\frac{x}{1}+\frac{2}{5}\right)$ OSATUGABEA

$$\frac{x^2}{2}+\frac{x}{60}=\frac{x^2}{3}+\frac{2x}{15}$$

$$\frac{30x^2}{60}+\frac{x}{60}=\frac{20x^2}{60}+\frac{8x}{60}$$

$$30x^2+x=20x^2+8x$$

$$30x^2-20x^2+x-8x=0$$

$$10x^2-7x=0$$

$$x(10x-7)=0 \rightarrow \boxed{x_1=0}$$

$$10x_2-7=0 \rightarrow 10x_2=7 \rightarrow \boxed{x_2=\frac{7}{10}}$$

h) $\frac{x}{3}\left(x-\frac{1}{20}\right)=\frac{x^2}{2}-\frac{1}{15}\left(2x-\frac{1}{2}\right)$ EZ DU SOLUZIORIK

$$\frac{x^2}{3}-\frac{x}{60}=\frac{x^2}{2}-\frac{2x}{15}+\frac{1}{30}$$

$$\frac{20x^2}{60}-\frac{x}{60}=\frac{30x^2}{60}-\frac{8x}{60}+\frac{2}{60}$$

$$20x^2-x=30x^2-8x+2$$

$$20x^2-x-30x^2+8x-2=0$$

$$-10x^2+7x-2=0 \quad \begin{cases} a=-10 \\ b=7 \\ c=-2 \end{cases}$$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{-7\pm\sqrt{7^2-4\cdot(-10)\cdot(-2)}}{2\cdot(-10)}=$$

$$=\frac{-7\pm\sqrt{49-80}}{-20}=\frac{-7\pm\sqrt{-31}}{-20}$$

EZ DU SOLUZIORIK

③ i) $\frac{x^2}{2} + \frac{x}{1} = \frac{2x^2-5}{3} - \frac{1}{1}$

$\frac{3x^2}{6} + \frac{6x}{6} = \frac{2(2x^2-5)}{6} - \frac{6}{6}$

$3x^2 + 6x = 4x^2 - 10 - 6$

$3x^2 + 6x = 4x^2 - 16$

$3x^2 + 6x - 4x^2 + 16 = 0$

$-x^2 + 6x + 16 = 0 \quad \begin{cases} a = -1 \\ b = 6 \\ c = 16 \end{cases}$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-6 \pm \sqrt{6^2 - 4.(-1).16}}{2.(-1)} = \frac{-6 \pm \sqrt{36+64}}{-2} =$

$= \frac{-6 \pm \sqrt{100}}{-2} = \frac{-6 \pm 10}{-2}$

$x_1 = \frac{-6+10}{2} = \frac{-4}{2} = -2$ $\boxed{x_1 = -2}$

$x_2 = \frac{-6-10}{-2} = \frac{-16}{-2} = 8$ $\boxed{x_2 = 8}$

⑥ Ebatzi:

a) $x^2 = 2x$

$x^2 - 2x = 0$

$x(x-2) = 0$ → $x = 0$; $x - 2 = 0$, $x = 2$

$\boxed{x_1 = 0}$

$\boxed{x_2 = 2}$

b) $x^2 = 9x - 8$

$x^2 - 9x + 8 = 0 \quad \begin{cases} a = 1 \\ b = -9 \\ c = 8 \end{cases}$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-9) \pm \sqrt{(-9)^2 - 4 \cdot 1 \cdot 8}}{2 \cdot 1} = \frac{9 \pm \sqrt{81 - 32}}{2} =$$

$$= \frac{9 \pm \sqrt{49}}{2} = \frac{9 \pm 7}{2} \rightarrow x_1 = \frac{9+7}{2} = \frac{16}{2} = 8 \quad x_2 = \frac{9-7}{2} = \frac{2}{2} = 1$$

$\boxed{x_1 = 8}$

$\boxed{x_2 = 1}$

c) $3x^2 = 4x$

$3x^2 - 4x = 0$

$x(3x - 4) = 0$ → $x = 0$; $3x - 4 = 0$, $3x = 4$, $x = \frac{4}{3}$

$\boxed{x_1 = 0}$

$\boxed{x_2 = 4/3}$

d) $2x^2 = 50$

$x^2 = 50 : 2$

$x^2 = 25$

$x = \sqrt{25}$

$x = \pm 5$ → $\boxed{x_1 = 5}$; $\boxed{x_2 = -5}$

6

e) $x^2 - 11x = -10$

$x^2 - 11x + 10 = 0 \quad \begin{cases} a = 1 \\ b = -11 \\ c = 10 \end{cases}$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-11) \pm \sqrt{(-11)^2 - 4 \cdot 1 \cdot 10}}{2 \cdot 1} =$$

$$= \frac{11 \pm \sqrt{121 - 40}}{2} = \frac{11 \pm \sqrt{81}}{2} = \frac{11 \pm 9}{2}$$

$x_1 = \frac{11+9}{2} = \frac{20}{2} = 10$

$x_2 = \frac{11-9}{2} = \frac{2}{2} = 1$

$\boxed{x_1 = 10}$ $\boxed{x_2 = 1}$

f) $-7x + 10 = -x^2$

$x^2 - 7x + 10 = 0 \quad \begin{cases} a = 1 \\ b = -7 \\ c = 10 \end{cases}$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-7) \pm \sqrt{(-7)^2 - 4 \cdot 1 \cdot 10}}{2 \cdot 1} =$$

$$= \frac{7 \pm \sqrt{49 - 40}}{2} = \frac{7 \pm \sqrt{9}}{2} = \frac{7 \pm 3}{2}$$

$x_1 = \frac{7+3}{2} = \frac{10}{2} = 5$ $\boxed{x_1 = 5}$

$x_2 = \frac{7-3}{2} = \frac{4}{2} = 2$ $\boxed{x_2 = 2}$

g) $7x = 2x^2$

$0 = 2x^2 - 7x$

$x(2x - 7) = 0$

$x = 0$

$2x - 7 = 0$

$2x = 7$

$x = 7/2$

$\boxed{x_1 = 0}$

$\boxed{x_2 = \frac{7}{2}}$

h) $24x^2 = 24$

$x^2 = 24 : 24$

$x^2 = 1$

$x = \sqrt{1} \rightarrow x = \pm 1$

$\boxed{x_1 = 1}$

$\boxed{x_2 = -1}$